

Personal Profile

I am a PhD at Monash University. My current research involves high frequency financial data analysis, Bayesian nonparametric methods and stochastic volatility modeling. Expertise in programming using Matlab and R. My education background equips me with great knowledge in probability theory, option pricing and machine learning.

Education

Monash University

Melbourne, Australia

Doctor of Philosophy

Jan 2023 - present

- Current research investigates the Bayesian nonparametric approach of detecting jumps in financial data with empirical application on option pricing.

University of Melbourne

Melbourne, Australia

Doctor of Philosophy

June 2022 - Jan 2023

- Transferred to Monash University.

University of Sydney

Sydney, Australia

Graduate Certificate in Economics Analysis

Aug 2021 - Mar 2022

- Major in Economics
- WAM: 86.5

Monash University

Melbourne, Australia

Bachelor of Commerce (1st Class Honors)

Feb 2017 - Dec 2020

- Major in Finance and Econometrics
- Graduated with High Distinction
- GPA: 3.9/4.0

Work Experience

Monash Business School

Melbourne, Australia

Teaching Associate

Feb 2023 - present

- High Dimensional Data Analysis (Head tutor)
- Data Analysis in Business (Excellent teaching team)
- Introductory Econometrics
- **Skills:** R programming, Teamwork, Time Management, Communication, Presentation skills.

Monash Business School

Melbourne, Australia

Research Assistant

Aug 2023 - Oct 2024

- Data Analysis for the Thailand Bond Market Association. Evaluate bond price jumps under the influence of regulation changes.

Melbourne Business School

Melbourne, Australia

Research Assistant

Aug 2022 - Jan 2022

- Use Python to extract high-frequency data of each stock in S&P 500 list from database - Refinitiv
- Using R to finish the data cleaning job, e.g. Extract the five-minute price for every stock and aggregate high frequency return to daily level.
- **Skills:** R programming, Python

Research Experience

Jump Modeling with Dirichlet Process Mixture

Melbourne, Australia

Monash University

June 2022 - Present

- Modelling extreme price movements – jumps, is crucial for risk management and option pricing in financial market and has been a challenging topic since jumps are difficult to detect together with the latent volatility movements. Jump diffusion models work as a complementary to volatility models, which accommodate fat tails of return distribution and the empirical implied volatility smile. We propose a Bayesian nonparametric method using Dirichlet process mixture to model jump. We investigate the robustness of this method under different jump specifications and verify its usefulness via option implied volatility and hedging strategies on S&P 500 returns.
- **Skills:** Matlab, Research skills, Report writing, Time management, Teamwork, Communication skills, Presentation skills

- Abstract: This paper explores the implications of producing forecast distributions that are optimized according to scoring rules that are relevant to financial risk management. We assess the predictive performance of optimal forecasts from potentially misspecified models for i) value-at-risk and expected shortfall predictions; and ii) prediction of the VIX volatility index for use in hedging strategies involving VIX futures. Our empirical results show that calibrating the predictive distribution using a score that rewards the accurate prediction of extreme returns improves the VaR and ES predictions. Tail-focused predictive distributions are also shown to yield better outcomes in hedging strategies using VIX futures.
- Link to the paper: <https://arxiv.org/abs/2303.01651>
- **Skills:** Matlab, R, Research skills, Report writing, Time management, Teamwork, Communication skills, Presentation skills

Honors & Awards

2024	Best Poster Runner Up , Time Series and Forecasting Symposium, Usyd
2023	Co-funded Monash Graduate Scholarship , Monash University
2022	Faculty of Business and Economics Graduate Research Scholarship , University of Melbourne
2021	2020 Dean's Honor (First Class) , Monash Business School
2020	Econometrics Memorial Scholarship , Department of Econometrics and Business Statistics, Monash
2020	Meritorious Winner , 2020 Interdisciplinary contest in Modelling